Our sound setup

Parts and budget | 16 October 2026

Goal: play and measure approximately 1.4–32 kHz tones, changing every 10 ms. Check the sound setup on the mouse rig first, then recalibrate for sheep.

What to get

Part and buying link	Why we need it	Qty	Cost
Earthworks M50	Measures above 32 kHz and has microphone-specific calibration data.	1	$1,299.00
Audient iD14 MKII	Works with Mac; plays sound and records the mic plus an electrical reference.	1	$349.00
Edifier MR3 powered pair	Affordable powered speaker with a specified range extending above 32 kHz.	1	$149.99
REED R8090	Checks the recording level scale using a known 1 kHz sound.	1	$219.00
Hosa HMIC-010 XLR 10 ft	Connects the microphone to the interface.	1	$17.99
Hosa HSS-005 TRS 5 ft	One cable for the speaker; one for the electrical reference.	2	$29.90
Gator RI-MICTP-FBM boom mic stand	Holds the microphone at a repeatable ear/reference position.	1	$34.99

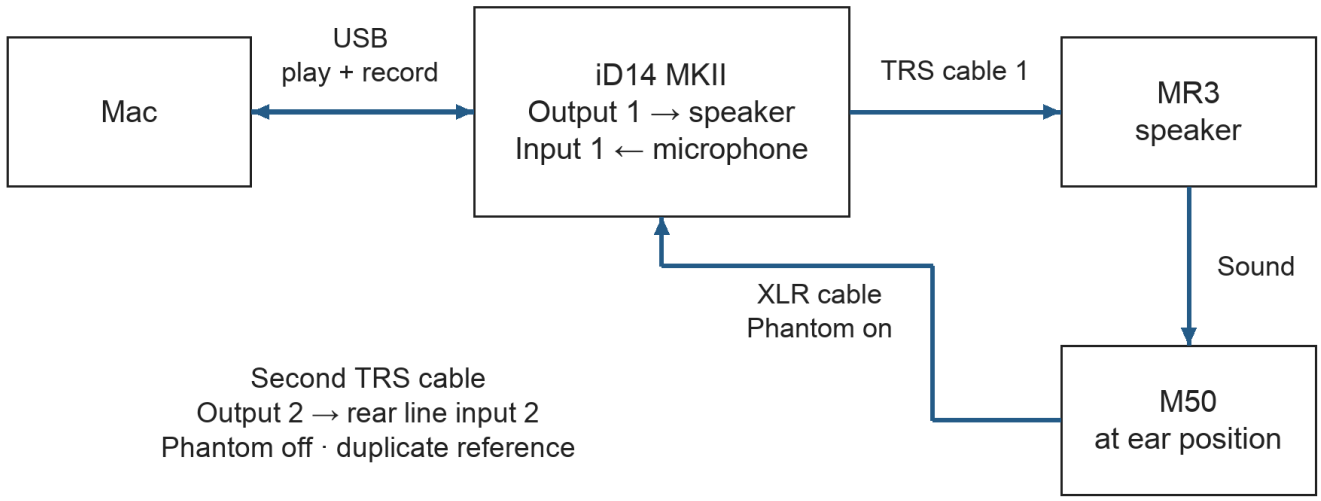
Parts total: $2,099.87. Plan for about $2,400 including estimated tax, shipping and small setup supplies.

Included: M50 clip and calibrator adapter, iD14 USB-C cable, and calibrator battery. Use the existing Mac and provide a stable speaker support.

The M50 is selected; iD14 is our recommended interface. MR3 still needs testing: its published range does not prove the required loudness or clean 10 ms changes at our chosen distance.

Prices checked 16 October 2026. Costs above are line totals; the two TRS cables cost $14.95 each. Buying links are in the part names.

Put it together



Connections for sound playback and recording

1. **Connect and position.** Connect as shown. Turn input monitoring off to prevent feedback; leave the front instrument input empty. Put the M50 tip at the ear/reference position and record distance and angle.
2. **Program the sequence.** Python builds one continuous waveform at 96 kHz: 960 samples per 10 ms pip, including a 2 ms ramp at each end. Send it to output 1 and a copy to output 2. Save the tone schedule and random seed.
3. **Calibrate.** Use the calibration file for that M50, checking coverage through 32 kHz. Use the R8090 at 94 dB to set the recording scale, then keep mic gain fixed. Remove the calibrator and adjust each tone's level using recordings at the ear position.
4. **Check the actual sound.** Record the microphone and electrical reference simultaneously during the actual 10 ms sequence. Check level at every frequency, short-tone transitions, ringing and stability through a full run. Check the reference channel's relative delay too. Repeat calibration when moving to sheep.

Before testing: the lab chooses distance, sound level and pass tolerances. The paper's approximately 80 dB sheep level is not automatically the mouse setting. Physical sound checks and later neural synchronization are still pending.

Specifications: M50 · iD14 · MR3 · R8090